INTERNATIONAL STATISTICAL ECOLOGY CONFERENCE 2027

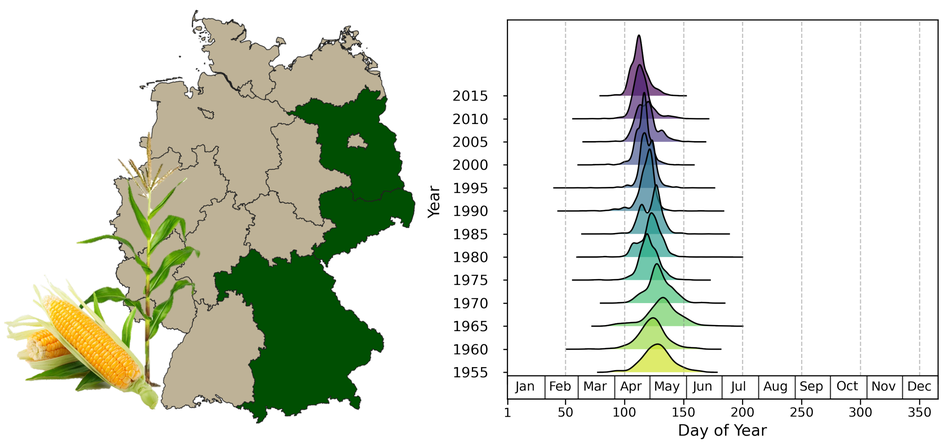
MODELING SOWING TIMING UNDER CLIMATE VARIABILITY USING RANDOM WALKS ON GRAPHS

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Introduction

The progression and duration of crop developmental stages are largely driven by environmental conditions [9]. Climate change is expected to alter the timing and duration of these stages [14, 13]. According to Baum *et al.* (2019) [2], sowing dates may account for up to 70% of maize yield variability. Depending on geographical location, maize sowing dates may remain unchanged or shift earlier or later in the year [10]. Climate change projections suggest that northern European regions, including Germany, may become increasingly suitable for maize cultivation [11]. In Germany, maize sowing dates have shifted towards earlier dates, as illustrated in Figure 1b. This shift has been associated with climate change and has intensified since the late 1980s [3]. Traditionally, phenological event timing is predicted using Growing Degree Days (GDD), an approach based primarily on air temperature [1]. In this context, we developed a probabilistic model based on four variables to predict maize sowing windows for each of the three German federal states considered in this study: Bavaria, Brandenburg, and Saxony, see Figure 1a. A region encompassing approximately 55% of the country's area devoted to major crops [4].



(a) Area of study. From top to bottom: Brandenburg, Saxony, and Bavaria.

(b) Distributions of maize sowing dates in Germany.

Figure 1: Geographical context and historical trends in maize sowing dates in Germany.

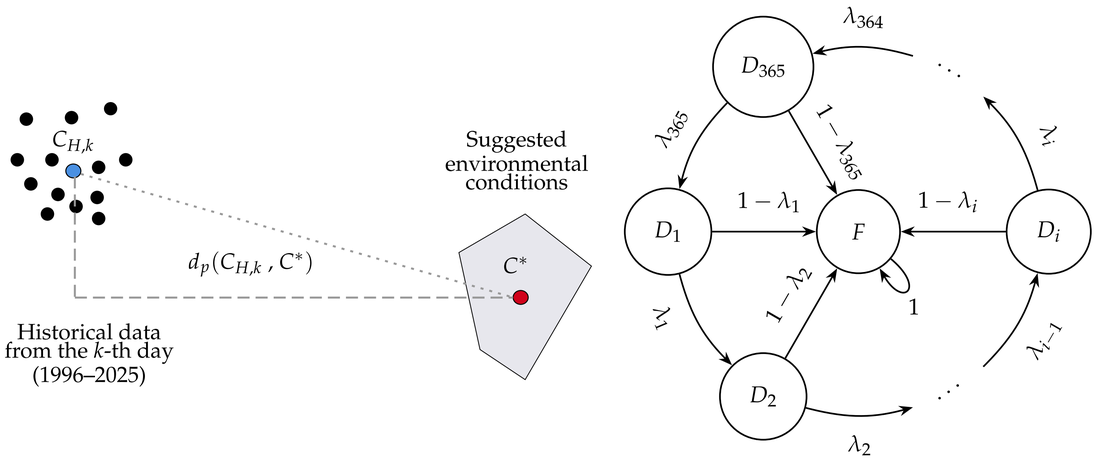
Methodology

Four raster datasets obtained from the Deutscher Wetterdienst (DWD) [8, 7, 6, 5] covering the period 1996–2025, were used as input data for the probabilistic model. These datasets provide daily measurements of mean air temperature, mean relative humidity, mean soil temperature at a depth of 5 cm, and mean soil moisture in the 0–10 cm soil layer, respectively, at a spatial resolution of 1×1 km.

Given $p \in \{1,2\}$, let $d(\cdot,\cdot) := d_p(\cdot,\cdot)$. Therefore, for each $k \in \{1,2,\ldots,365\}$

$$\lambda_k = 1 - \frac{d(C_{H,k}, C^*)}{\max\limits_{l \in \{1,\ldots,365\}} \{d(C_{H,l}, C^*)\}}. \tag{1}$$

denotes the transition probability.



(a) p-value of Kolmogorov-Smirnov test at different population sizes.

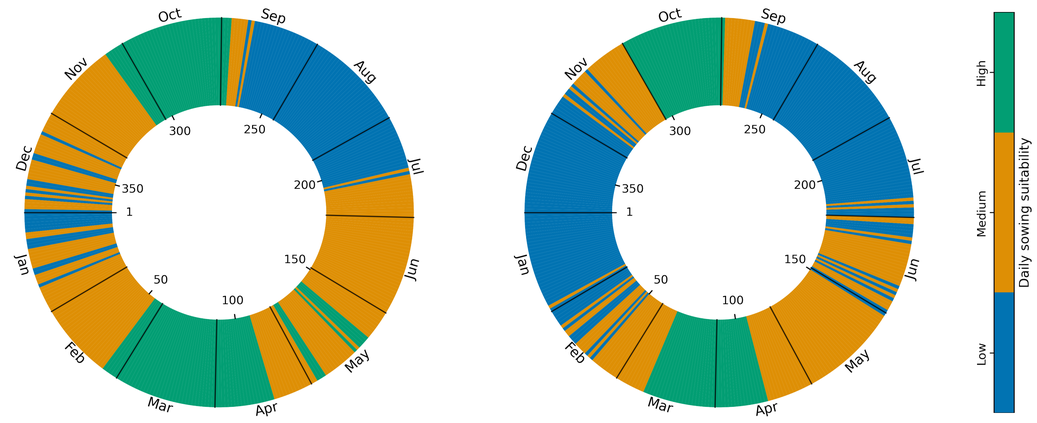
(b) χ^2 goodness-of-fit for Luria-Delbruck distribution.

Figure 2: Statistical comparisons.

Agroclimatic variable	Interval used in this study
Mean air temperature	11.10 ± 2.51 °C
Mean air relative humidity	65–85 %
Mean soil temperature at a depth of 5 cm	8–10 °C
Mean soil moisture in the 0–10 cm soil layer	80–90 %

Table 1: Daily

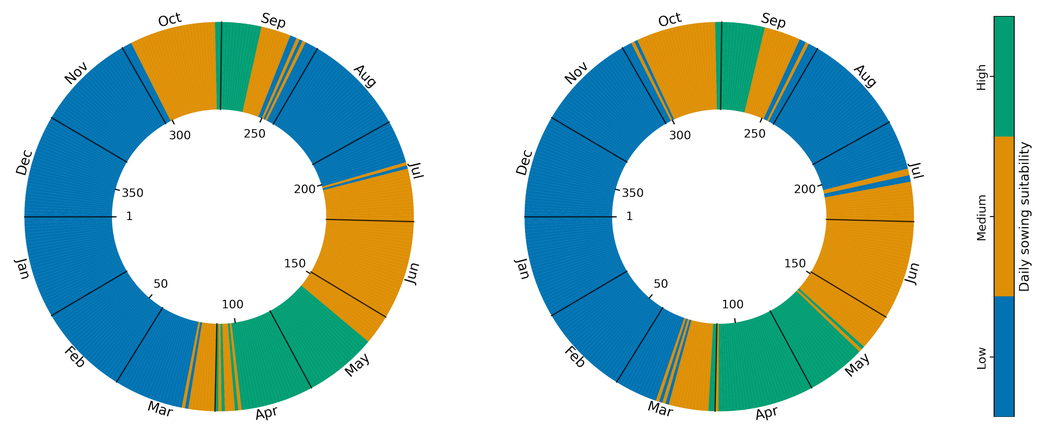
Results



(a) p-value of Kolmogorov-Smirnov test at different population sizes.

(b) χ^2 goodness-of-fit for Luria-Delbruck distribution.

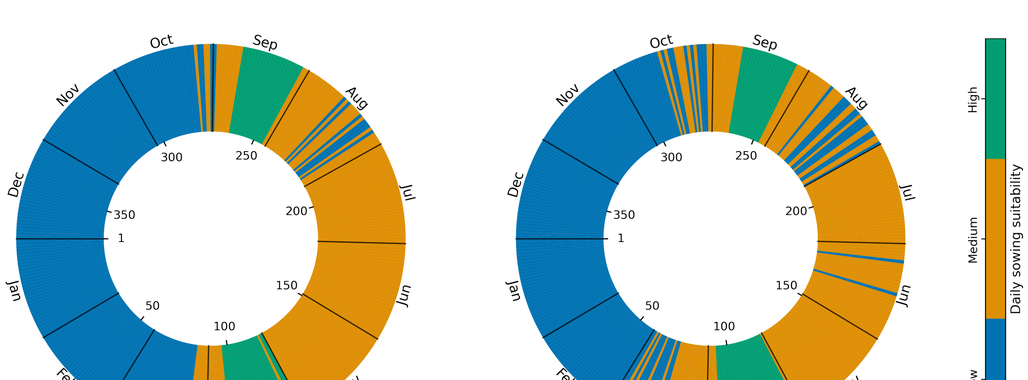
Figure 3: Statistical comparisons.



(a) p-value of Kolmogorov-Smirnov test at different population sizes.

(b) χ^2 goodness-of-fit for Luria-Delbruck distribution.

Figure 4: Statistical comparisons.



(a) p-value of Kolmogorov-Smirnov test at different population sizes.

(b) χ^2 goodness-of-fit for Luria-Delbruck distribution.

Figure 5: Statistical comparisons.

Conclusions

References

[1] M. Amare and B. Balana. Climate change, income sources, crop mix, and input use decisions: Evidence from nigeria. *Ecological Economics*, 211:107892, 2023.

[2] M. Baum, S. Archontoulis, and M. Licht. Planting date, hybrid maturity, and weather effects on maize yield and crop stage. *Agronomy Journal*, 111(1):303–313, 2019.

[3] F. Chmielewski, A. Müller, and E. Bruns. Climate changes and trends in phenology of fruit trees and field crops in germany, 1961–2000. *Agricultural and Forest Meteorology*, 121(1):69–78, 2004.

[4] C. Duden, C. Nacke, and F. Offermann. German yield and area data for 11 crops from 1979 to 2021 at a harmonized spatial resolution of 397 districts. *Scientific Data*, 11(1), Jan. 2024.

[5] DWD Climate Data Center (CDC). Daily grids of mean soil moisture under maize for germany, version v1.0, 2024. Accessed August 5, 2026, at `https://opendata.dwd.de/climate_environment/CDC/grids_germany/daily/soil_moisture/maize/`.

[6] DWD Climate Data Center (CDC). Daily grids of soil temperature at 5 cm depth, 2025. Accessed July 29, 2026, at `https://opendata.dwd.de/climate_environment/CDC/grids_germany/daily/soil_temperature_5cm/`.

[7] DWD Climate Data Center (CDC). Raster data set of mean relative humidity in % for Germany - HYRAS-DE-HURS v6-1, Version v6-1, 2025. Accessed July 31, 2026, at `https://opendata.dwd.de/climate_environment/CDC/grids_germany/daily/hyras_de/humidity/`.

[8] DWD Climate Data Center (CDC). Raster data set of mean temperature in ° C for Germany - HYRAS-DE-TAS v6-1, Version v6-1, 2025. Accessed July 31, 2026, at `https://opendata.dwd.de/climate_environment/CDC/grids_germany/daily/hyras_de/air_temperature_mean/`.

[9] Z. Liu. *Crop Phenology: Timing is everything*, page 257–294. Springer Nature Singapore, 2026.

[10] S. Mason, T. Galusha, and Z. Kmail. Planting date influence on yield of drought-tolerant maize with different maturity classifications. *Agronomy Journal*, 110(1):293–299, 2018.

[11] T. Miedaner and P. Juroszek. Global warming and increasing maize cultivation demand comprehensive efforts in disease and insect resistance breeding in north-western europe. *Plant Pathology*, 70(5):1032–1046, 2021.

[12] J. Olesen, C. Børgesen, L. Elsgaard, T. Palosuo, R. Rötter, A. Skjelvåg, P. Peltonen-Sainio, T. Börjesson, M. Trnka, F. Ewert, S. Siebert, N. Brisson, J. Eitzinger, E. van Asselt, M. Oberforster, and H. van der Fels-Klerx. Changes in time of sowing, flowering and maturity of cereals in europe under climate change. *Food Additives & Contaminants: Part A*, 29(10):1527–1542, Aug. 2012.

[13] R. Patil, M. Lægdsman, J. Olesen, and J. Porter. Growth and yield response of winter wheat to soil warming and rainfall patterns. *The Journal of Agricultural Science*, 148(5):553–566, 2010.

[14] P. Peltonen-Sainio, A. Kangas, Y. Salo, and L. Jauhiainen. Grain number dominates grain weight in temperate cereal yield determination: Evidence based on 30 years of multi-location trials. *Field Crops Research*, 100(2-3):179–188, Feb. 2007.